Aarti Barve

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Summary

AI/ML Engineer with 5+ years of experience in software engineering, AI, and Machine Learning, with a focused specialization in Generative AI, NLP, and large language models. Improved mobile app search and user engagement through LLM-based search, RAG systems, and semantic modeling. Strong proficiency in data manipulation and preprocessing cleansing large datasets, handling missing values, feature engineering, and ensuring data integrity using Python, Pandas, and NumPy. Experienced in building AI solutions and data workflows at scale, covering ETL, feature engineering, and cloud-based pipelines.

Technical Skills

Programming & Libraries: Python, R, SQL, TypeScript, ReactJS

Databases & Tools: PostgreSQL, MySQL, Oracle, MongoDB, Graph DB, Pandas, NumPy

Tools & Frameworks: FastAPI, GitHub, Node.js, Docker, Jenkins, Tableau, PowerBI, Streamlit

AI/ML & Cloud: Machine Learning Models, TensorFlow, PyTorch, LangChain, LangGraph, CrewAI, HuggingFace Models, LLaMA 3, AWS (Bedrock, Sagemaker), GCP (Vertex AI, BigQuery, AutoML, Looker/LookML), MCP, Prompt Engineering, Guardrails, MLOps, OWASP Top 10 for Application Security, Microservices, Kubernetes, AI Agents

Experience

AI/ML Engineer Jan 2026 – Present
PNC Bank *Phoenix, AZ*

- Contributed to developing an AI-powered credit risk and decisioning platform that combined real-time data with ML models and LLM-based explanations, helping automate loan approvals and risk profiling while giving underwriters clearer reasoning behind each decision
- Built production-grade LangChain pipelines integrating custom tools, function-calling, and retrieval chains to orchestrate LLM interactions across structured and unstructured knowledge bases, reducing hallucination through grounded context; tracked groundedness score, tool-call success rate, and latency
- Designed LangGraph-powered agentic systems with memory persistence, checkpointing, and retry logic for fault-tolerant, resumable multi-step workflows aligned with enterprise reliability and compliance requirements
- Built automated LLM evaluation frameworks (BERTScore, custom rubric-based evaluators) and integrated evaluation score trends into MLflow dashboards for centralized model monitoring, usage tracking, and degradation alerting
- Built FastAPI services for real-time risk scoring, AI decisioning, and agent orchestration, plus the data layer for internal reporting dashboards; tracked p95 latency, throughput, and error rate
- Built stakeholder-facing dashboards in ReactJS to track and visualize KPIs and performance metrics for the credit decisioning platform, giving risk and business leaders real-time visibility into model performance and operational throughput
- Ensured compliance with banking regulations, model governance, explainability standards, and enterprise data security policies

AI/ML Engineer Mar 2025 – Jan 2026
CDreams Foundation *Phoenix, AZ*

- Developed an enterprise knowledge assistant using Claude 3.5 Sonnet to automate document analysis, contextual Q&A, and support workflows; tracked deflection rate, answer accuracy, user satisfaction, and cost per query
- Performed ad hoc data analysis using Python and SQL to validate pipeline outputs, identify retrieval gaps, and deliver actionable insights to business stakeholders, ensuring data accuracy and integrity across knowledge base workflows
- Built RAG pipelines using LangChain, LlamaIndex, and Pinecone over internal knowledge bases, policy documents, and business manuals; tracked Recall@k, faithfulness, and hallucination rate
- Implemented prompt engineering and prompt optimization techniques to improve response accuracy, reasoning quality, and domain-specific performance of LLM-based applications
- Designed semantic search and embedding pipelines using vector search, document chunking, and metadata filtering to improve knowledge retrieval performance

GenAI Engineer Oct 2024 – Mar 2025
American Express *Phoenix, AZ*

- Built a GenAI-powered search pipelines combining semantic, fuzzy, and keyword-based retrieval to serve 60K+ daily queries across US, UK, AU, and Japan markets, boosting click-through rate by 15% and digital containment by 3%
- Conducted quantitative analysis using Python (Pandas) and SQL to monitor pipeline health, validate search result accuracy, and perform root cause analysis on query failures, maintaining data integrity across 60K+ daily transactions
- Built and maintained operational dashboards in Tableau to visualize search performance KPIs (CTR, containment rate, query volume) across four markets, enabling leadership to track trends and make data-driven decisions
- Benchmarked Evaluation Framework for semantic and fuzzy search models using 1,200 GPT-labeled queries. Incorporated human-in-the-loop feedback to fine-tune thresholds, significantly improving model precision and search relevance
- Integrated event-driven data pipelines using Kafka to enable real-time query processing between search services
- Developed a high-precision NER pipeline (regex, spaCy, NLTK) to interpret dates, transaction amounts, and merchant names; tracked entity extraction F1 and empty-query resolution rate, resolving 3,000+ empty queries per month
- Designed the high-level architecture for the NER pipeline and integrated it into the existing system using a microservices based approach to ensure scalability and modular deployment
- Improved relevance of AMEX mobile in-app search by analyzing user logs, and using LLMs to generate FAQ content and tailored results, boosting query alignment reducing need for chat agent escalation
- Contributed to building the Typeahead API that improved search functionality in the American Express mobile app, handling 45K+ requests per day across the US

- Created a documentation portal in Docusaurus using ReactJS and TypeScript, streamlining onboarding and serving as a live knowledge base for improveing cross-team alignment and delivered insights to stakeholder questions

Data Scientist Jan 2024 – Jun 2024
ZigZag Venture Capital *Irvine, CA*

- Applied BERTopic, LDA, and transformer-based NLP models to analyze user-generated startup ideas and extract thematic trends, helping stakeholders evaluate and refine early-stage business ideas
- Developed a predictive framework using model-driven scoring to analyze user submissions, enabling product managers to prioritize high-potential initiatives and achieve a 2x higher investment success rate
- Automated executive dashboard generation and KPI reporting using Tableau and Power BI, improving visibility into business idea, industry trends and evaluation metrics while reducing manual reporting efforts by 50%
- Partnered with cross-functional stakeholders product managers, analysts, and executives, to define data requirements, validate scoring models, and align on evaluation criteria for strategic initiatives
- Ensured data accuracy and model integrity through validation checks and QA processes on user submission datasets, maintaining consistency across scoring pipelines and executive reporting outputs

Senior Software Engineer Mar 2021 – Apr 2023
Infosys Limited *Pune, India*

- Developed and deployed Java-based automation tools to extract and format high-volume financial datasets from ERP platforms, reducing processing time by 40% and increasing reporting accuracy across audit and compliance tasks
- Designed and implemented a microservices-based architecture across multiple business units, enabling service modularity and scalability; integrated ActiveMQ for reliable asynchronous messaging and inter-service communication
- Built and maintained automated enterprise-level eCommerce integration workflows using MuleSoft APIs and Jenkins CI/CD pipelines, streamlining deployments and reducing manual handoffs
- Partnered with business and finance stakeholders to translate reporting requirements into SQL data models, ETL logic, and automated workflows, driving alignment between technical deliverables and operational priorities
- Optimized existing SQL queries and ETL pipelines for performance and scalability, reducing processing time by 40% and improving dashboard load times for downstream reporting consumers
- Implemented automated data validation and QA checks within batch processing workflows to proactively detect anomalies, ensure data integrity, and reduce manual reconciliation efforts
- Implemented ETL pipelines and batch processing workflows to aggregate, transform, and process data from multiple sources, including integration with third-party services for data enrichment and validation
- Managed MySQL/PostgreSQL databases with a focus on schema design, query optimization, and performance tuning; improved application efficiency using Redis caching and ensured reliability

Projects

Marketing & Sales AI Agent — *Snowflake Cortex AI, Dynamic Tables*

- Built an end-to-end AI agent within Snowflake using Cortex AI (Cortex Analyst & Cortex Search) to enrich and analyze structured and unstructured data; implemented automated pipelines using Dynamic Tables for sentiment analysis and classification, and designed a semantic layer with enterprise-grade security (masking policies) to deliver scalable, business-ready insights

Voice Assistant — *LiveKit, ElevenLabs, GPT-4o-mini*

- Optimized an LLM-to-TTS pipeline by implementing dynamic turn detection and response streaming, cutting the perceived end-to-end response time from 0.98s to 0.34s for a more natural, conversational user experience

AI Agents for Content Generation — *CrewAI, Multi-Agent System, GPT-4o-mini*

- Developed a multi-agent content creation system using CrewAI and LLaMA 2 7B, coordinating Planner, Writer, and Editor agents to automate ideation, drafting, and refinement enabling scalable and repeatable content workflows

Agentic Document Q&A System with MCP — *LangChain, GPT, RAG*

- Built an MCP-powered, multi-agent document Q&A system where Planner, Retriever, and Answer agents communicate via MCP to generate accurate responses: the Planner identifies query intent, the Retriever searches semantic embeddings in a vector database, and the Answer agent synthesizes context from retrieved documents, enabling dynamic, context-aware, and grounded answers over large research paper collections

Education

University of California Irvine Irvine, CA
Master of Science in Business Analytics *Aug 2024*

Dr. Babasaheb Ambedkar Marathwada University India
Bachelor of Engineering in Computer Science *Jun 2020*

Certifications

- LangSmith Essentials - LangChain Academy
- LangGraph Essentials - Python - LangChain Academy
- Google Cloud Certified Generative AI Leader
- Fundamentals of Agents – Hugging Face
- Engineer Data for Predictive Modeling with BigQuery ML Skill Badge – ETL, Dataflow, Data Pipelines
- Machine Learning Specialization – ML Models, Hyperparameter Tuning
- Build LookML Objects in Looker Skill Badge - Data Modeling